USN | | | | | | | | | | [22CSEDE21]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

DATA MODELLING WITH NOSQL DATABASE

Time: 3 Hours **Maximum Marks: 100**

		PART-A	**5 x2=10**		
1.	**a.**	Define "Impedance Mismatch" in the context of relational databases and object-oriented applications.	L2	CO1	**02**
	b.	Why is sharding useful in distributed databases?	L1	CO2	**02**
	c.	Name one key feature of MongoDB that supports high availability.	L1	CO4	**02**
	d.	What does eventual consistency mean in the context of Cassandra?	L2	CO5	**02**
	e.	Illustrate the main Key features of Neo4j.	L3	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the role of ACID properties in relational databases and how they support data consistency.	L2	CO1	**05**
		OR			
	b.	Summarize the benefits and challenges of using schema-less databases in applications with frequently changing requirements.	L4	CO1	**05**
3.	**a.**	Explain the trade-off between consistency and availability in key-value databases.	L2	CO2	**05**
		OR			
	b.	How does the combination of sharding and replication in distributed database systems improve scalability and fault tolerance?	L2	CO2	**05**
4.	**a.**	Explain how CRUD operations are performed in MongoDB.	L2	CO3	**05**
		OR			
	b.	Identify scenarios where MongoDB is not the best choice for data storage.	L4	CO3	**05**
5.	**a.**	Demonstrate the key characteristics of column-family data stores.	L3	CO4	**05**
		OR			
	b.	Describe the transaction model in Cassandra DB and its implications for data integrity.	L4	CO4	**05**
6.	**a.**	Describe some capabilities of Cypher in Neo4j.	L2	CO5	**05**
		OR			
	b.	Discuss scenarios where Neo4j might not be the best fit for an application.	L3	CO5	**05**
7.	**a.**	Explain the consistency model used in Neo4j DB and how it affects data operations.	L2	CO6	**05**
		OR			
	b.	Discuss the implications of aggregate orientation in NoSQL databases, providing an example.	L3	CO2	**05**

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		PART-C	6 X 10=60		
8.	a.	Design a NoSQL data model for a social media application to store user profiles, posts, and comments. Explain your data model, aggregation choices, and how it supports data access.	L3	CO1	10
		OR			
	b.	Explain how materialized views work in NoSQL databases and how they can improve query performance. Give an example of a scenario where a materialized view would be beneficial.	L3	CO1	10
9.	a.	Discuss the flexibility and limitations of data structure in key-value databases, and provide an example of how this affects application design.	L3	CO2	10
		OR			
	b.	Create a data model for a key-value store to manage a user's shopping cart in an e-commerce application. Explain your choice of key structure, value format, and strategies for scaling and consistency.	L3	CO2	10
10.	a.	Design a data model for a blogging platform using MongoDB, detailing the structure of documents and relationships among collections. Explain how this model supports common operations.	L3	CO3	10
		OR			
	b.	Explain the significance of the aggregation framework in MongoDB, detailing its components and common use cases. Provide an example of an aggregation pipeline.	L2	CO3	10
11.	a.	Design a data model for a social media application using Cassandra DB, detailing the structure of column families and their relationships. Explain how this model supports common operations like CRUD.	L3	CO4	10
		OR			
	b.	Discuss scenarios where Cassandra might not be the best fit for an application.	L2	CO4	10
12.	a.	Discuss the scaling capabilities of Neo4j, including its query features and suitable use cases. Also, mention when Neo4j might not be the best choice.	L2	CO5	10
		OR			
	b.	Design a data model for a travel application using Neo4j, detailing the structure of nodes, relationships, and properties. Explain how this model supports common operations.	L3	CO5	10
13.	a.	What strategies can be used to optimize query performance in Neo4j, and how can these techniques be applied to improve efficiency in a large-scale graph database?	L3	CO6	10
		OR			
	b.	Evaluate the advantages and disadvantages of using Neo4j for a recommendation engine in an e-commerce application.	L3	CO6	10